

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

1. Answer the following short questions:

(6x5=30)

What are generalized coordinates? Identify generalized coordinates and the constraints of a plane pendulum. Use D'Alembert's principle to obtain the equation of motion of plane pendulum.

A bee goes out from its hive in a spiral path given in plane polar coordinates

$$r = b \exp(kt), \quad \theta = ct$$

where b, k, c are positive constants. Show that the angle between the velocity vector and the acceleration vector remains constant as the bee moves outward.

If L is a Lagrangian for a system of n degrees of freedom satisfying Lagrange equation of motion, show by direct substitution that

$$L' = L + \frac{d}{dt} F(q_1, \dots, q_n; t),$$

What are cyclic coordinates? Given a set of cyclic coordinates, express the Lagrange's equation of motion in terms of Routhian function.

Show that the transformation

$$\begin{aligned} Q &= q \tan p \\ P &= \ln(\sin p) \end{aligned}$$

is canonical transformation.

Prove that the Poisson bracket of two constants of motion is itself a constant even when the constants depend on time explicitly.

2. Answer the following questions.

(3x10=30)

Consider the motion of a particle in a central force field

$$V(r) = -\frac{k}{r}$$

Write down the Lagrangian in polar coordinates and integrate the equation of motion to derive

$$\theta(r) = \int \frac{l \, dr}{r^2 \sqrt{2\mu \left(E + \frac{k}{r} - \frac{l^2}{2\mu r^2} \right)}} = \text{constant},$$

where E is the total energy and l is the angular momentum. Now change variables as $u = \frac{1}{r}$ to derive the equation of a conic section

$$\frac{a}{r} = 1 + e \cos \theta.$$

2. Consider a one parameter family of transformations

$$q_i(t) \rightarrow Q_i(s, t) \quad s \in \mathbb{R}$$

such that $Q_i(0, t) = q_i(t)$. Show that if the Lagrangian is invariant under this transformation, then there exists a conserved quantity (Noether Theorem).

Define action functional. State Hamilton's principle of least action. Derive Lagrange equation of motion by using Hamilton's principle of least action.

Generalized Coordinates:

In classical mechanics, generalized coordinates are a set of independent coordinates that completely describe the configuration of a system. They are chosen such that they simplify the problem by reducing the number of coordinates needed to describe the system's motion. For example, instead of using Cartesian coordinates (x, y, z) to describe the position of each particle in a system, you might choose a different set of coordinates that are more convenient for the problem at hand.

Identifying Generalized Coordinates and Constraints of a Plane Pendulum:

Consider a simple pendulum swinging in a plane. We can describe its configuration using two generalized coordinates:

1. **Angular displacement (θ):** The angle the pendulum makes with the vertical axis.
2. **Arc length (s):** The length of the arc along the path of the pendulum bob.

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2. **Arc length (s):** The length of the arc along the path of the pendulum bob.

The constraint of the pendulum is that the length of the pendulum remains constant.

Mathematically, this constraint can be expressed as $s = l\theta$, where l is the length of the pendulum.

Using d'Alembert's Principle:

d'Alembert's principle states that the virtual work done by the sum of the applied forces and the inertial forces, when subjected to virtual displacements consistent with the constraints of the system, is zero.

For a pendulum, the forces acting on it are gravity and tension. Using d'Alembert's principle, we can write the equation of motion for the pendulum:

1. **Gravity force:** $-mgl \sin(\theta)$, where m is the mass of the pendulum bob, g is the acceleration due to gravity, and l is the length of the pendulum.

For a pendulum, the forces acting on it are gravity and tension. Using d'Alembert's principle, we can write the equation of motion for the pendulum:

1. **Gravity force:** $-mgl \sin(\theta)$, where m is the mass of the pendulum bob, g is the acceleration due to gravity, and l is the length of the pendulum.
2. **Tension force:** T , which acts along the direction of the pendulum.

Now, applying d'Alembert's principle, we equate the virtual work of these forces to zero.

Obtaining the Equation of Motion:

The equation of motion for the pendulum can be derived by considering the torque about the pivot point. The torque is the moment of force, and it equals the rate of change of angular momentum.

The torque acting on the pendulum due to the gravitational force is $-mgl \sin(\theta)$, and it is equal to the time derivative of the angular momentum $\downarrow$, which is $I \frac{d^2\theta}{dt^2}$, where I is the moment of inertia of the pendulum bob.

The torque acting on the pendulum due to the gravitational force is $-mgl \sin(\theta)$, and it is equal to the time derivative of the angular momentum, which is $I \frac{d^2\theta}{dt^2}$, where I is the moment of inertia of the pendulum bob.

Setting these equal, we get:

$$-mgl \sin(\theta) = I \frac{d^2\theta}{dt^2}$$

This is the equation of motion for the plane pendulum, describing its angular acceleration as a function of time.

Conclusion:

In summary, we've identified the generalized coordinates (θ and s) and constraints of the plane pendulum, applied d'Alembert's principle to obtain the equation of motion, and derived the equation describing the pendulum's angular acceleration. Understanding these concepts and their applications is crucial in classical mechanics and physics in general.

Date: ___/___/20

Day: _____

$$= -\frac{\partial \dot{B}}{\partial q} \frac{\partial A}{\partial p} + \frac{\partial A}{\partial q} \frac{\partial \dot{B}}{\partial p} + \frac{\partial \dot{A}}{\partial q} \frac{\partial B}{\partial p} - \frac{\partial B}{\partial q} \frac{\partial \dot{A}}{\partial p}$$

$$= \left(\frac{\partial A}{\partial q} \frac{\partial \dot{B}}{\partial p} + \frac{\partial \dot{A}}{\partial q} \frac{\partial B}{\partial p} \right) - \left(\frac{\partial \dot{B}}{\partial q} \frac{\partial A}{\partial p} + \frac{\partial A}{\partial p} \frac{\partial \dot{B}}{\partial q} \right)$$

$$= \frac{\partial}{\partial t} \left(\frac{\partial A}{\partial q} \frac{\partial B}{\partial p} \right) - \frac{\partial}{\partial t} \left(\frac{\partial A}{\partial p} \frac{\partial B}{\partial q} \right)$$

$$= \frac{\partial}{\partial t} \left(\frac{\partial A}{\partial q} \frac{\partial B}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial B}{\partial q} \right)$$

$$\{B, A\} + \{A, B\} = \frac{\partial}{\partial t} \{A, B\}$$

Now equa (3) becomes

$$\{A, B\} + \frac{\partial}{\partial t} \{A, B\} = 0$$

Shows $\{A, B\}$ is constant of motion.

* * * *

2)

$$r = be^{kp(kt)}, \quad \theta = ct$$

The equation of velocity in polar coordinates

$$\vec{v} = \dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta \quad \text{--- (1)}$$

$$\dot{r} = \frac{d}{dt} r = \frac{d}{dt} be^{kt} = bke^{kt} \quad \text{--- (2)}$$

$$\dot{\theta} = \frac{d}{dt} \theta = \frac{d}{dt} ct = c \quad \text{--- (3)}$$

put (2) and (3) in (1)

$$\vec{v} = bke^{kt} \hat{e}_r + bce^{kt} \hat{e}_\theta \quad (4)$$

Now equa of acceleration in polar coordinates.

$$\vec{a}(t) = (\ddot{r} - r\dot{\theta}^2) \hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \hat{e}_\theta \quad (5)$$

$$\ddot{r} = \frac{d}{dt} \dot{r} = \frac{d}{dt} bke^{kt} = bk^2 e^{kt} \quad (6)$$

$$\ddot{\theta} = \frac{d}{dt} \dot{\theta} = \frac{d}{dt} c = 0 \quad (7)$$

put (6) and (7) in (5)

$$\vec{a}(t) = (bk^2 e^{kt} - be^{kt} c^2) \hat{e}_r + (0 + 2bcke^{kt}) \hat{e}_\theta$$

$$= b(k^2 - c^2) e^{kt} \hat{e}_r + 2bcke^{kt} \hat{e}_\theta$$

Application for the law of dot product for the velocity and acceleration vector

$$\vec{a} \cdot \vec{v} = av \cos \phi$$

$$\cos \phi = \frac{\vec{a} \cdot \vec{v}}{av}$$

$$\vec{a} \cdot \vec{v} = b^2 k e^{2kt} (k^2 - c^2) + 2b^2 c^2 e^{2kt}$$

Date: / / 20 Day:

$$\vec{a} \cdot \vec{v} = [(k^2 - c^2) + 2c^2] b^2 k e^{2kt}$$

$$a = |\vec{a}| = \sqrt{[b(k^2 - c^2) e^{kt}]^2 + [2bck e^{kt}]^2}$$

$$a = b e^{kt} \sqrt{(k^2 - c^2)^2 + 4c^2 k^2}$$

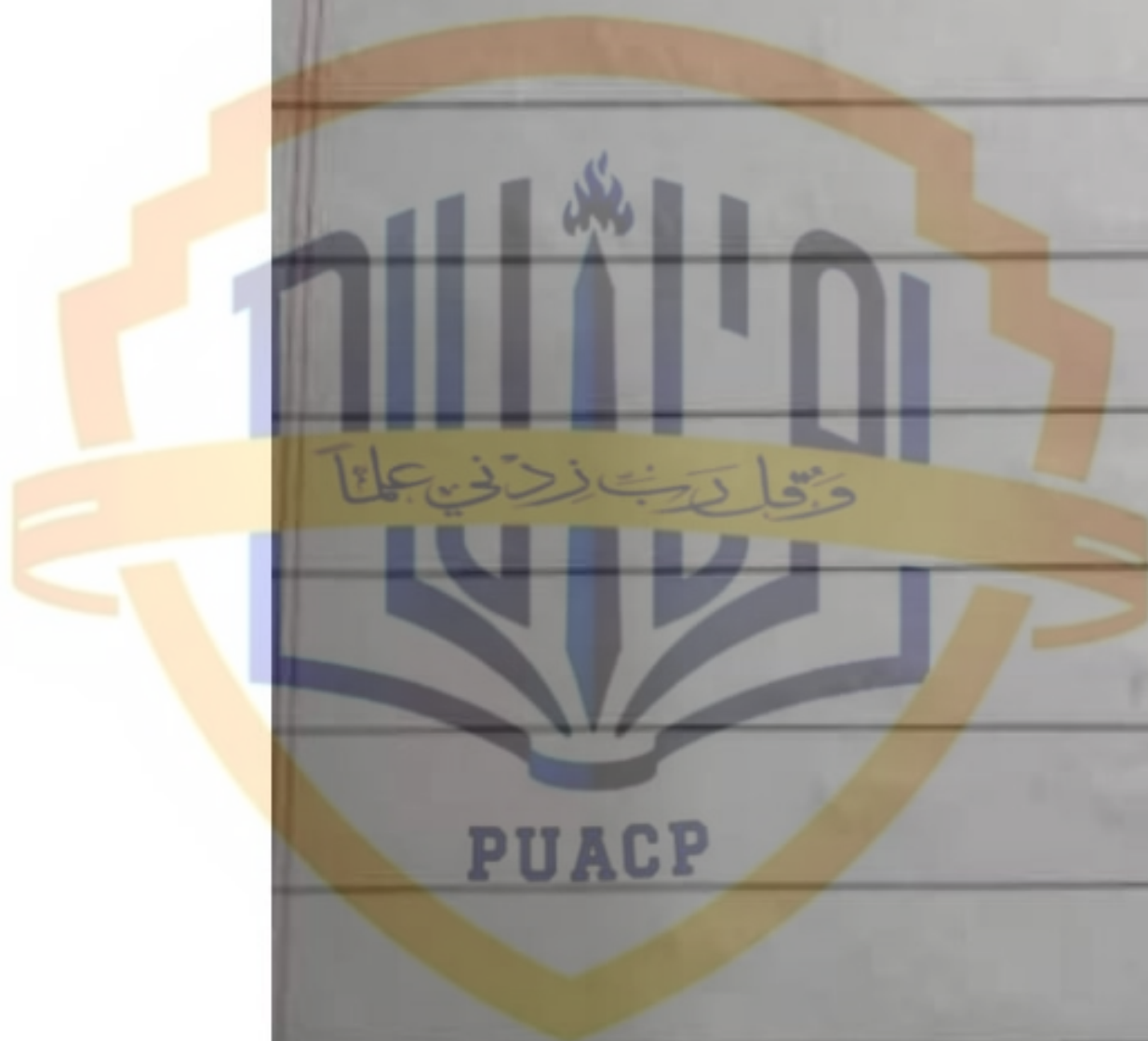
$$v = |\vec{v}| = \sqrt{(b k e^{kt})^2 + (b c e^{kt})^2}$$

$$= b e^{kt} \sqrt{k^2 + c^2}$$

Now

$$\cos \phi = \frac{[(k^2 - c^2) + 2c^2] b^2 k e^{2kt}}{b e^{kt} \sqrt{(k^2 - c^2)^2 + 4c^2 k^2} \cdot b e^{kt} \sqrt{k^2 + c^2}}$$

= constant.



Problem 02

imp ✓

If L is a Lagrangian for a system of n degrees of freedom satisfying Lagrange's equation of motion, show by direct substitution that

$$L' = L + \frac{d}{dt}F(q_1, q_2, \dots, q_n, t) = L + \dot{F}$$

Also satisfies the Lagrange's equations of motion where F is an arbitrary differentiable function of its argument.

Sol:

Lagrange's Equation for n Degrees of Freedom

If L' is another Lagrangian for a system then Lagrange's equation of motion becomes

$$\frac{d}{dt} \left(\frac{\partial L'}{\partial \dot{q}_i} \right) - \frac{\partial L'}{\partial q_i} = 0$$

Substituting value of L'

$$\frac{d}{dt} \left(\frac{\partial (L + \dot{F})}{\partial \dot{q}_i} \right) - \frac{\partial (L + \dot{F})}{\partial q_i} = 0$$

$$\left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} \right] + \left[\frac{d}{dt} \left(\frac{\partial \dot{F}}{\partial \dot{q}_i} \right) - \frac{\partial \dot{F}}{\partial q_i} \right] = 0$$

As

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

So

$$\frac{d}{dt} \left(\frac{\partial \dot{F}}{\partial \dot{q}_i} \right) - \frac{\partial \dot{F}}{\partial q_i} = 0 \quad \text{----- (1)}$$

Force is given as

$$F = F(q_1, q_2, \dots, q_n, t)$$

$$\frac{dF}{dt} = \frac{\partial F}{\partial q_1} \frac{\partial q_1}{\partial t} + \frac{\partial F}{\partial q_2} \frac{\partial q_2}{\partial t} + \dots + \frac{\partial F}{\partial q_n} \frac{\partial q_n}{\partial t} + \frac{\partial F}{\partial t} \frac{\partial t}{\partial t}$$

$$= \frac{\partial F}{\partial q_1} \dot{q}_1 + \frac{\partial F}{\partial q_2} \dot{q}_2 + \dots + \frac{\partial F}{\partial q_n} \dot{q}_n + \frac{\partial F}{\partial t}$$

$$\dot{F} = \sum_i \frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial t}$$

$$\frac{\partial \dot{F}}{\partial \dot{q}_i} = \frac{\partial F}{\partial q_i} (1) + 0$$

$$\frac{\partial \dot{F}}{\partial \dot{q}_i} = \frac{\partial F}{\partial q_i}$$

Hence

$$\frac{d}{dt} \left(\frac{\partial \dot{F}}{\partial \dot{q}_i} \right) = \frac{d}{dt} \left(\frac{\partial F}{\partial q_i} \right) = \frac{\partial \dot{F}}{\partial q_i} \quad \text{----- (2)}$$

Which is the same as eq(1). Hence the result is proved.

Problem 22

PUACP

Q.6 What are cyclic coordinates? Also show that a coordinate cyclic in L is also cyclic in H . [PU 03, 04]

Cyclic Coordinates

If only the time derivative of a particular coordinate appears in the Lagrangian, but not the coordinate itself, then the coordinate is called cyclic or ignorable coordinate.

For example, the Lagrangian for a free particle is

$$L = \frac{1}{2} m \dot{r}^2$$

It involves only time derivative of coordinate ($\dot{r}$) but not r itself. Thus this coordinate is ignorable. A coordinate cyclic in L is also cyclic in H .

Q.7 Express Lagrange equations of motion in terms of Routhian function. [PU 10, 11, 12, 15, 16, GUJ 16]

procedure

Routh's Procedure

The cyclic coordinates can be removed from the Lagrangian by a transformation suggested by Routh. Routh's method is similar to the method for obtaining Hamilton's equations of motion, except that the Legendre transformation of the Lagrangian is applied only to the cyclic coordinates.

Consider a Lagrangian consists of k cyclic coordinates $\{q_c\}$ and $n-k$ non-cyclic coordinates $\{q_i\}$ then

$$L = L(q_i, \dot{q}_i, \dot{q}_c; t)$$

Now define a new Lagrangian that depends only on the non-cyclic coordinates and the conserved momenta by a Legendre transform to eliminate the cyclic coordinates.

$$R(q_i, \dot{q}_i, p_c; t) = L(q_i, \dot{q}_i, \dot{q}_c; t) - p_c \dot{q}_c$$

Where R is reduced Lagrangian called Routhian.

Taking differential of

$$\frac{\partial R}{\partial q_i} dq_i + \frac{\partial R}{\partial \dot{q}_i} d\dot{q}_i + \frac{\partial R}{\partial p_c} dp_c + \frac{\partial R}{\partial t} dt = \frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i + \frac{\partial L}{\partial \dot{q}_c} d\dot{q}_c + \frac{\partial L}{\partial t} dt - p_c d\dot{q}_c - \dot{q}_c dp_c$$

Using Lagrangian equations

$$\frac{\partial R}{\partial q_i} dq_i + \frac{\partial R}{\partial \dot{q}_i} d\dot{q}_i + \frac{\partial R}{\partial p_c} dp_c + \frac{\partial R}{\partial t} dt = \dot{p}_i dq_i + p_i d\dot{q}_i + p_c d\dot{q}_c + \frac{\partial L}{\partial t} dt - p_c d\dot{q}_c - \dot{q}_c dp_c$$

$$\frac{\partial R}{\partial q_i} dq_i + \frac{\partial R}{\partial \dot{q}_i} d\dot{q}_i + \frac{\partial R}{\partial p_c} dp_c + \frac{\partial R}{\partial t} dt = \dot{p}_i dq_i + p_i d\dot{q}_i + \frac{\partial L}{\partial t} dt - \dot{q}_c dp_c$$

Comparing coefficients of dq_i and $d\dot{q}_i$ on both sides

$$\left. \begin{aligned} \frac{\partial R}{\partial q_i} &= \dot{p}_i \\ \frac{\partial R}{\partial \dot{q}_i} &= p_i \end{aligned} \right\} \dots(1)$$

From set of equations (1)

$$\frac{\partial R}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial R}{\partial \dot{q}_i} \right) = \dot{p}_i - \dot{p}_i$$

$$\frac{\partial R}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial R}{\partial \dot{q}_i} \right) = 0$$

Routhian behaves like a Lagrangian for the noncyclic coordinates and like a Hamiltonian for the cyclic coordinates.

$$p dq - P dQ = d\left(\frac{q}{Q}\right)$$

which is an exact differential. So the generating function is q/Q .

Show that the transformations $Q = q \tan p$ and $P = \ln \sin p$ is canonical.

$$\text{--- Taking } p dq - P dQ$$

$$\text{---} = p dq - \ln \sin p \, d(q \tan p)$$

$$\text{---} = p dq - \ln \sin p \left(q \sec^2 p \, dp + \tan p \, dq \right)$$

$$\text{---} = p dq - q \ln |\sin p| \sec^2 p \, dp - \tan p \, dq \ln \sin p$$

$$\text{---} = (p - \tan p) dq - q \sec^2 p \ln |\sin p| dp$$

$$\text{---} = p dq - \tan p \ln \sin p \, dq - q \ln |\sin p| \sec^2 p \, dp$$

$$\text{---} = (p - \tan p \ln \sin p) dq + q (-\ln |\sin p| \sec^2 p) dp$$

$$\text{---} = \left\{ q(-p - \tan p \ln \sin p) \right\}$$

which is an exact differential.

Thus, $[F, [G, K]] + [G, [K, F]] + [K, [F, G]] = 0$

which proves the Jacobi's identity.

Ex. 5. Show that the Poisson bracket of two constants of motion is itself a constant of motion. (Garwal 1995; Agra 81)

Solution : In Jacobi's identity, we put $K = H$, then

$$[F, [G, H]] + [G, [H, F]] + [H, [F, G]] = 0$$

Now, if F and G are constants of motion, then $[F, H] = 0$ and $[G, H] = 0$

Therefore, $[H, [F, G]] = 0$

which means that the dynamic variable $[F, G]$ is constant of motion. Thus the Poisson bracket of two constants of motion is itself a constant of motion.

under canonical transformation.

Example 4.19:- Consider the motion of particle in repulsive coulomb potential,

$$V(r) = -\frac{k}{r}$$

Write down the Lagrangian in polar coordinates and integrate the equation of motion to derive

$$\theta(r) = \int \frac{l dr}{r^2 \sqrt{2\mu \left(E + \frac{k}{r} - \frac{l^2}{2\mu r^2} \right)}} + \text{constant}$$

where E is total energy and l is angular momentum.

Now change variables $u = \frac{1}{r}$ to derive equation of conic as

$$\frac{p}{r} = 1 + e \cos \theta$$

Solution:- Total energy of two body problem is,

$$E = \frac{1}{2} \mu \dot{r}^2 + \frac{l^2}{2\mu r^2} + V(r) \Rightarrow \dot{r}^2 = \frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2} \Rightarrow \dot{r} = \sqrt{\frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2}}$$

$$\Rightarrow \frac{dr}{dt} = \sqrt{\frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2}} \Rightarrow dt = \frac{dr}{\sqrt{\frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2}}}$$

Since $\dot{\theta} = \frac{l}{\mu r^2}$, $\frac{d\theta}{dt} = \frac{l}{\mu r^2}$ gives,

$$d\theta = \frac{l}{\mu r^2} dt \Rightarrow d\theta = \frac{l}{\mu r^2} \frac{dr}{\sqrt{\frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2}}} \Rightarrow \theta = \int \frac{\frac{1}{r^2} dr}{\sqrt{\frac{\mu^2}{l^2} \left\{ \frac{2}{\mu} (E - V) - \frac{l^2}{\mu^2 r^2} \right\}}}$$

$$\Rightarrow \theta = \int \frac{\frac{1}{r^2} dr}{\sqrt{\frac{2\mu E}{l^2} - \frac{2\mu V}{l^2} - \frac{1}{r^2}}}$$

$$\Rightarrow \theta = \int \frac{\frac{1}{r^2} dr}{\sqrt{\frac{2\mu E}{l^2} + \frac{2\mu k}{l^2 r} - \frac{1}{r^2}}}$$

Let $r = \frac{1}{u}$, $dr = -\frac{1}{u^2} du$

$$\theta = \int \frac{-\frac{1}{u^2} du}{\sqrt{\frac{2\mu E}{l^2} + \frac{2\mu k}{l^2} u - u^2}}$$

Comparing above equation with

$$\Rightarrow \theta = \int \frac{-du}{\sqrt{\frac{2\mu E}{l^2} + \frac{2\mu k}{l^2} u - u^2}}$$

$$\int \frac{dx}{\sqrt{a + bx + cx^2}} = \frac{1}{\sqrt{-c}} \cos^{-1} \left(\frac{-b - 2cx}{\sqrt{b^2 - 4ac}} \right)$$

we find that,

$$\frac{2\mu E}{l^2} = a, \quad \frac{2k\mu}{l^2} = b, \quad c = -1, \quad x = u$$

$$\theta = \cos^{-1} \left\{ \frac{-\left(\frac{2k\mu}{l^2}\right) + 2u}{\sqrt{\left(\frac{2k\mu}{l^2}\right)^2 + 4\frac{2\mu E}{l^2}}} \right\} \Rightarrow \cos \theta = \frac{-\left(\frac{2k\mu}{l^2}\right) \left[1 - \frac{l^2 u}{\mu k}\right]}{\left(\frac{2k\mu}{l^2}\right) \sqrt{1 + \frac{2El^2}{\mu k^2}}}$$

$$\Rightarrow \sqrt{1 + \frac{2El^2}{\mu k^2}} \cos \theta = -1 + \frac{l^2 u}{\mu k} \Rightarrow r = \frac{\frac{l^2 u}{\mu k}}{1 + \sqrt{1 + \frac{2El^2}{\mu k^2}} \cos \theta}$$

$$\Rightarrow r = \frac{p}{1 + e \cos \theta} \Rightarrow \frac{p}{r} = 1 + e \cos \theta$$

where $\frac{l^2}{\mu k} = p$ and $e = \sqrt{1 + \frac{2El^2}{\mu k^2}}$

formation then angular momentum is conserved.

FINALLY

Space is homogeneous $\rightarrow$ Linear momentum conserved

Time is homogeneous $\rightarrow$ Total Energy conserved

Isotropy of space $\rightarrow$ Angular momentum conserved

✓ NOETHER'S THEOREM

Statement: This theorem states that every symmetry transformation is associated with some conserved quantity.

PROOF: Consider a one parameter family of transformation

$$Q_i(t) \longrightarrow Q_i(s, t) \quad s \in \mathbb{R} \quad \text{--- (1)}$$

where s is parameter of transformation. This type of transformation is called Continuous transformation.

This Continuous transformation is said to be the symmetry of the system if the Lagrangian of the system does not depend upon the parameter s .

Statement #2 This implies that

$$\frac{\partial L}{\partial s} [Q_i(s, t), \dot{Q}_i(s, t), t] = 0 \quad \text{--- (2)}$$

for each invariance of Lagrangian function under a continuous transformation there is a conservation theorem. //

PUACP

(24)

Consider $L = L(Q_i(s, t), \dot{Q}_i(s, t), t)$

$$\frac{\partial L}{\partial s} = \frac{\partial L}{\partial Q_i} \frac{\partial Q_i}{\partial s} + \frac{\partial L}{\partial \dot{Q}_i} \frac{\partial \dot{Q}_i}{\partial s} + \frac{\partial L}{\partial t} \frac{\partial t}{\partial s}$$

$$\frac{\partial L}{\partial s} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}_i} \right) \frac{\partial Q_i}{\partial s} + \frac{\partial L}{\partial \dot{Q}_i} \frac{\partial \dot{Q}_i}{\partial s} \leftarrow \left(\frac{\partial t}{\partial s} = 0 \right) \quad (26)$$

Now taking

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{Q}_i} \frac{\partial Q_i}{\partial s} \right] = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}_i} \right) \frac{\partial Q_i}{\partial s} + \frac{\partial L}{\partial \dot{Q}_i} \frac{d}{dt} \left(\frac{\partial Q_i}{\partial s} \right)$$

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{Q}_i} \frac{\partial Q_i}{\partial s} \right] = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}_i} \right) \frac{\partial Q_i}{\partial s} + \frac{\partial L}{\partial \dot{Q}_i} \frac{\partial \dot{Q}_i}{\partial s} \quad (27)$$

Comparing (26) and (27)

$$\frac{\partial L}{\partial s} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}_i} \frac{\partial Q_i}{\partial s} \right)$$

The expression is true for any value of s , so let $s=0$

then $\frac{\partial L}{\partial s} \big|_{s=0} = 0$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}_i} \frac{\partial Q_i}{\partial s} \right) \bigg|_{s=0} = 0$$

$$\frac{d}{dt} h = 0 \Rightarrow h \text{ is conserved}$$

So $h = \frac{\partial L}{\partial \dot{Q}_i} \frac{\partial Q_i}{\partial s}$

$h = p_i \frac{\partial Q_i}{\partial s}$ is Required Conserved

The more general result is

$$h_j = \sum_{i=1}^n p_i \frac{\partial Q_i}{\partial s_j} \bigg|_{\{s_j\}=0} \quad (28)$$

are the Conserved Quantities.

So the minimum condition is Euler Lagrange Equation.

Derive Lagrange Equation of least action principle.

Consider a dynamical system moves from its configuration 1 at time t_1 to another configuration 2 at time t_2 in such a way that

$$I[q(t)] = \int_{t_1}^{t_2} dt L(q(t), \dot{q}(t), t)$$

has the least value for the correct path of motion.

The correct path of motion is that for which the action (or functional) is stationary.

$$\text{Taking } \delta I[q] = \int_{t_1}^{t_2} dt \delta L(q, \dot{q}, t)$$

$$\delta I[q] = \int_{t_1}^{t_2} dt \left[\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} + \frac{\partial L}{\partial t} \delta t \right]$$

$$\delta I[q] = \int_{t_1}^{t_2} dt \left[\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] + \frac{\partial L}{\partial t} \delta t \quad (15)$$

For least variation $\delta I[q] = 0$

$$\text{Taking } \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \delta q \right) = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q + \left(\frac{\partial L}{\partial \dot{q}} \right) \frac{d}{dt} \delta q$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \delta q \right) = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q}$$

Integrating both sides w.r.t. t from t_1 to t_2

$$\int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \delta q \right) dt = \int_{t_1}^{t_2} \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] dt$$

$$\left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2} = \int_{t_1}^{t_2} \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] dt$$

$$\left. \frac{\partial L}{\partial \dot{q}_i} \cdot \delta q_i \right|_{t_1}^{t_2} = \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \delta q_i dt + \int_{t_1}^{t_2} \frac{\partial L}{\partial q_i} \delta q_i dt$$

Since $\delta q_i(t_1) = 0 = \delta q_i(t_2)$ Hence LHS = 0

So $\int_{t_1}^{t_2} \frac{\partial L}{\partial q_i} \delta q_i dt = - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \delta q_i dt$ — (15)

Rearranging (15) we get

$$\delta I[q_i] = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q_i} \delta q_i \right) dt + \int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i dt$$

Substituting (16) in above equation

$$\delta I[q_i] = \int_{t_1}^{t_2} \frac{\partial L}{\partial q_i} \delta q_i dt - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \delta q_i dt$$

$$\delta I[q_i] = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right] \delta q_i dt$$

but $\delta I[q_i] = 0$ So

$$\int_{t_1}^{t_2} dt \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right] \delta q_i = 0$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \quad \text{--- (17)}$$

which is Lagrange Equation of motion

Every information about a dynamical system is contained in Lagrangian. $L = L(q_i, \dot{q}_i, t)$

If q_i is cyclic, Then Lagrangian does not depend upon q_i

So $\frac{\partial L}{\partial q_i} = 0$

Then from (17) $\frac{\partial L}{\partial \dot{q}_i} = \text{Conserved}$

We know that $\frac{\partial L}{\partial \dot{q}_i} = p_i$ (Generalised Momenta)

The set of generalized coordinates is given by

⇒ FUNCTIONAL: A functional denoted by $I[y]$ is a function of function $y (= f(x))$. In Calculus of variation our task is so to find this $y (= f(x))$ for which $I[y]$ has maximum or minimum value.

So $I[y] = \int_a^b f[\underline{y(x)}, \underline{y'(x)}, x] dx \quad \text{--- (1)}$

